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Back-Imaging of Pressure-Sensitive Paint to Determine Close Proximity Ground Effects of Propellers

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The ground effects of propellers have been extensively studied in the literature both theoretically and experimentally via a variety of methods; however, there are limitations with regards to global analyses of the mean and unsteady flow along the surface of the ground plate. This study provides a proof of concept that back-imaged pressure-sensitive paint (PSP) can successfully analyze the mean and unsteady surface pressure distributions along such a ground plate. In particular, the unsteady nature of the pressure field was successfully measured utilizing this technique. By analyzing the pressure field in the frequency domain, the unsteady characteristics of the flow were able to be quantitatively determined. In particular, the PSP-derived power spectral density of the surface pressure fluctuations show primary peaks at a frequency which agrees both with the theoretical predicted blade rotation and the spectral results calculated from the supplementary Kulite pressure sensors. The amplitude of this primary spectral peak is then shown globally at every PSP pixel location, which showed a clear low-amplitude region near the center of the blade radius, and a clear maximum-amplitude ring located at approximately 75% of the blade radius. Lastly, the concavity of this spectral amplitude profile changed signs near the blade-radius percent where previous works by the authors qualitatively measured a dividing streamline between primarily tangential flow and radial flow. These data provide a promising area for potential use of PSP to more fully understand the flowfield under rotor blades in ground effect, thereby improving safety and performance.

I. Nomenclature

AA-PSP	=	Anodized Aluminum Pressure-Sensitive Paint
CFD	=	Computational Fluid Dynamics
I	=	Intensity (of PSP emission)
$\frac{h}{d}$	=	Ratio of propeller height above ground to diameter of propeller
N	=	Frame
p	=	Pressure (kPa)
p'	=	Fluctuating Component of Pressure (kPa)
$\bar{p}$	=	Mean Pressure (kPa)
PC-PSP	=	Polymer-Ceramic Pressure-Sensitive Paint
PSP	=	Pressure-Sensitive Paint

Sub/superscripts

kul	=	denotes quantity measured by discrete pressure transducers (Kulites)
PSP	=	denotes quantity measured by PSP
ref	=	denotes reference quantity

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II. Introduction

The ground effect on a spinning rotor blade is defined as where an impermeable planar surface (i.e., ground) is parallel and in close proximity to the spinning rotor, and has been studied theoretically [1–3] and experimentally [2–7] for decades. The ground offers two benefits to the rotor: an increase in thrust produced, and a decrease in power required, for a given rotational speed, which both contribute to the propulsive efficiency augmentation [8]. These benefits also come with a complicated flowfield observed on the ground plate, including a center tangential flow and an outer radial flow regime [8, 9]. With the emerging use of quadcopters and applications of air taxis, studies on rotor and propeller ground effect have garnered more attention in recent years [8–12]. Ground effect on multi-rotor aerial vehicles brings additional complexity to the flowfield where additional fountain flow and recirculation fountain flow are observed when placing the rotor within two diameters of the ground in hover [10]. Understanding the flowfield on the ground plate will benefit the operation of the modern air taxi takeoff and landing as the rotor outwash in the ground effect creates safety concerns for humans near the landing zone [13]. Avoiding the area where fountain flow and strong radial flow occur will help ensure the safe operation of the multi-rotor drone and air-taxi vehicles.

Visualizing and measuring this flowfield's interaction at the ground has been completed in past research through several methods. Dekker et al. [10] utilized volumetric particle image velocimetry to study the flowfield around rotors in ground effect scenarios, allowing a visualization of this flowfield from a side view. Cai et al. [9], meanwhile, qualitatively analyzed the flowfield at the ground plate using three techniques: surface tuft flow visualization that revealed ground plate surface flow direction and stagnation points, smoke visualization, and surface oil flow visualization. These techniques allowed various aspects of the flow to be visualized. Cai et al. [9] combined these measurements with discrete surface pressure transducer (Kulite) data, allowing for global qualitative pressure information to be successfully deduced [9]. However, this method was inherently limited by the need to correlate discrete pressure measurements to the global flowfield. Although these techniques remain useful, their inability to provide global quantitative information of the flowfield is limiting. To capture unsteady phenomenon or provide a direct, quantitative analysis of the entire flowfield near the ground plate, a global surface pressure measurement technique is required.

Pressure-sensitive paint (PSP) represents a potential method to fill this requirement. PSP is a global surface pressure sensor applied to a model. The paint contains a luminophore chemical that responds to the partial pressure of oxygen. When illuminated by a specific wavelength of light, this luminophore emits another wavelength of light at an intensity inversely related to the local partial pressure of oxygen. This emission can be captured by a camera, recording the intensity of the PSP at each pixel in the image. Knowing the relationship between pressure and this intensity via a calibration, the pressure at each pixel in the image can therefore be calculated. This provides a global pressure field over the test object [14, 15]. PSP is often utilized in wind-tunnel testing and as verification for computational fluid dynamics (CFD) models [14]. It has potential applications in the case of small rotorcraft ground interaction when applied to the ground plate under the propeller, theoretically allowing the pressure at each location on the ground to be determined over time.

Fast PSP is a subset of PSP designed for unsteady aerodynamic measurements. Traditional PSP can be relatively slow to respond to changes in pressure relative to other techniques, such as discrete sensors. Fast PSP is altered to allow the oxygen to more easily interact with the luminophore in the paint, reducing the response time (and increasing the frequency response). This allows the paint to more quickly react to changes in the pressure field, allowing unsteady aerodynamic phenomena to be captured on the order of ones of kHz [15]. There are two main types of fast PSP: anodized aluminum (AA-PSP) and polymer-ceramic (PC-PSP). AA-PSP uses a porous, anodized aluminum model that increases the surface area and aids bonding of the luminophore [15]. PC-PSP includes small particles of ceramic, either mixed with the paint or applied before the luminophore solvent. This also increases the surface area and porosity of the paint, allowing oxygen to reach the luminophore with less difficulty; therefore, decreasing the response time [15]. PC-PSP is utilized in this study to quantitatively capture the unsteady phenomenon that occurs under the propeller on a global scale.

PSP relies on the surface of the model being in sight of the camera during data collection. The propeller, however, partially blocks the surface of the ground from being viewed during ground-effect testing; therefore, back-imaging of the PSP was attempted in the current study. In this modification of traditional PSP, the luminophore mixture was applied to a clear substrate. This material was then illuminated and imaged from the opposite side of the PSP layer, allowing the PSP to be utilized without visual access to the surface of the test object, as discussed in Ref. [16]. This technique has been attempted occasionally in the literature, and the results suggest that back-imaging and illuminating PSP produces comparable results to traditional imaging [16]. However, the technique is still relatively novel. In this study, back-imaging and back-illumination was utilized as the most effective way to avoid visual interference from the propeller.

III. Experimental Setup

The experimental setup is shown in Fig. 1. The propeller is attached to a frame on one side of the ground plate. The ground plate is a 1/4 in. cast acrylic sheet. This ground plate is coated with PSP on the side of the propeller (right in Fig. 1, Side B). On the opposing side, a blue laser is utilized to illuminate the PSP while a high speed camera records images (Side A). The physical setup is shown in Fig. 2

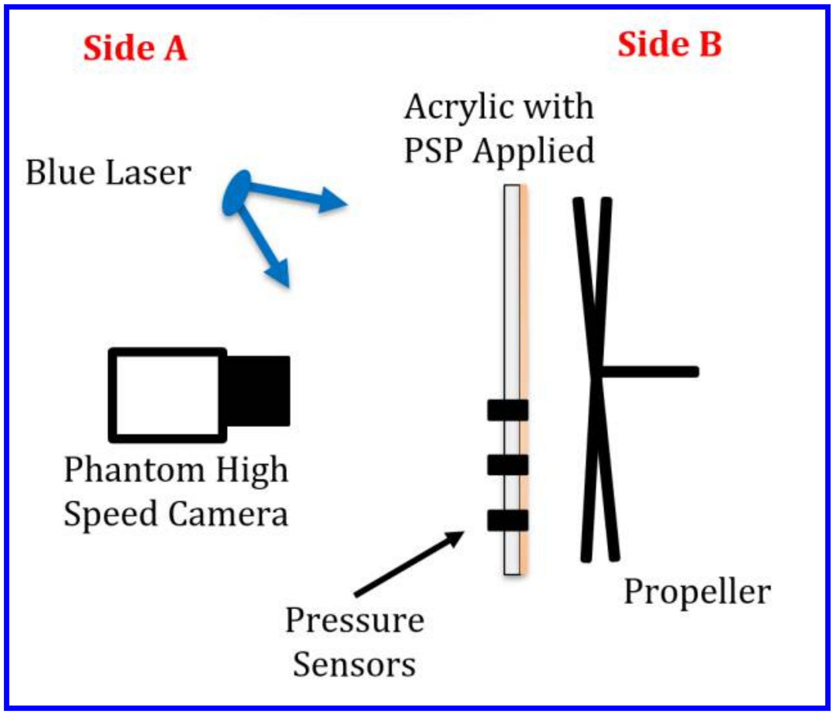

Fig. 1 Schematic showing PSP and rotor blade ground-effect experimental setup.

A. Propeller

The propeller utilized in this experiment is an APC 17 x 7 "thin electric" propeller [17]. It is rotated by an E-Flight Power 60-400 kV brushless outrunner electic motor and PSW 30-108 constant-voltage power supply to maintain a speed of approximately 60 revolutions per second, based on previous measurements. The propeller is kept at a constant 1.7 inches from the ground plate, creating an $\frac{h}{d}$ of 0.1, where $\frac{h}{d}$ is the ratio of h, the distance between the propeller and the ground plate, to d, the diameter of the propeller. In this experiment, the propeller speed and distance from the ground plate are constant.

B. PSP

The PSP utilized is a recipe derived from Sakaue et al. [18] and was made in-house. The ingredients and concentrations are shown in Table 1.

Table 1 Composition of in-house PC-PSP utilized, derived from Sakaue et al. [18]; the total amount is scaled as needed

Ingredient	Concentration	Function
Ru(dpp)	0.1 mM	Luminophore
RTV rubber	2 g	Binder
Silica particles	3 g	Ceramic Particle
Dichloromethane	50 ml	Solvent

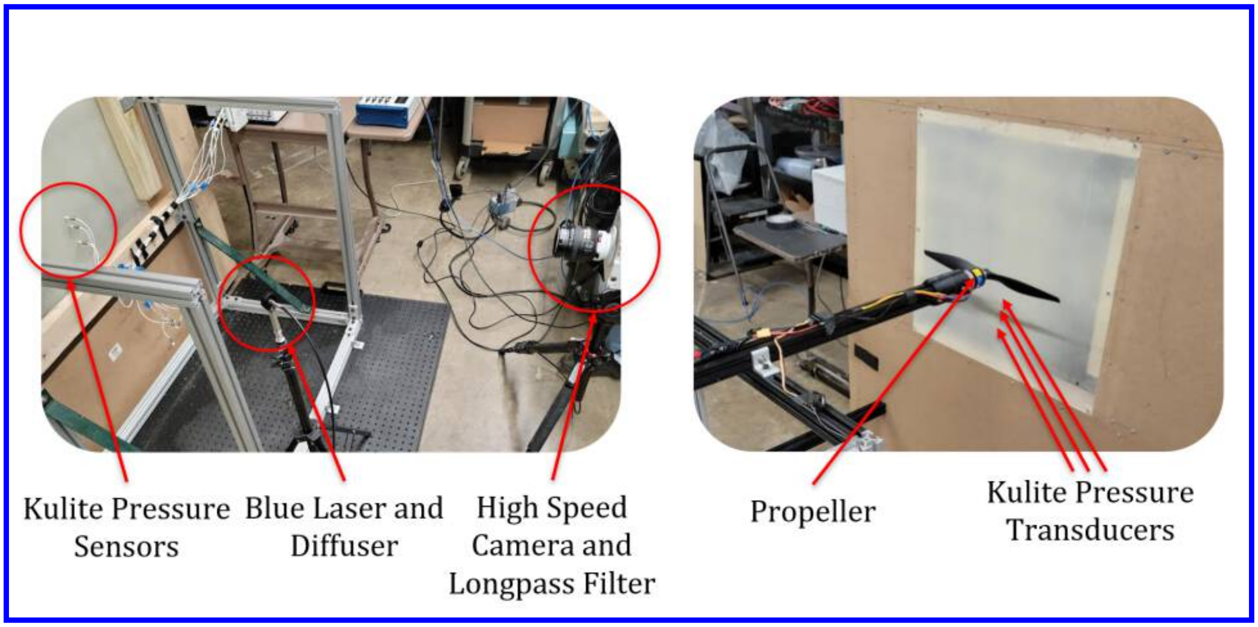

Fig. 2 Annotated images of the experimental setup, the left corresponding to Side A of Fig. 1.

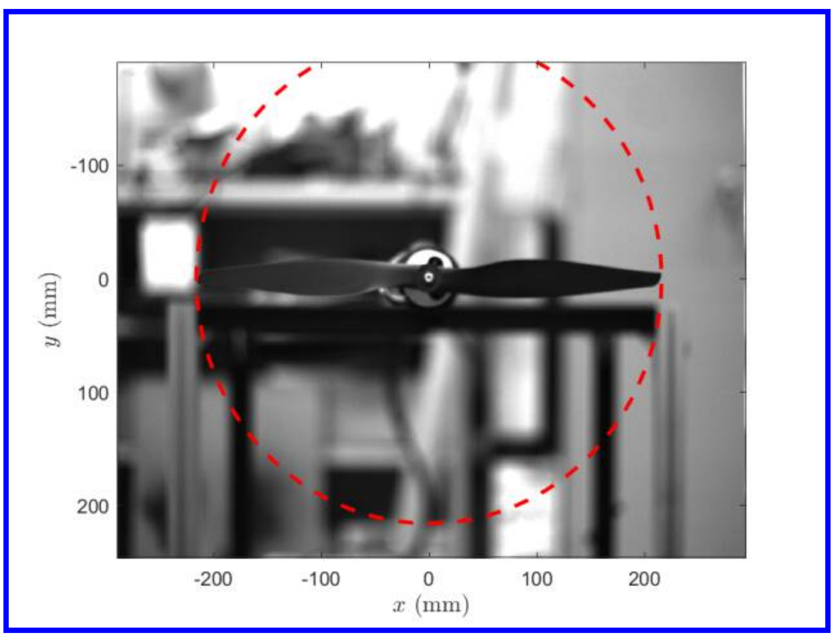

Fig. 3 Example field of view of the camera. In this image, the acrylic ground plate and PSP are removed to show the propeller location.

The PSP is mixed thoroughly utilizing a stirring plate and ultrasonic water bath. The mixture is then applied to the cast acrylic sheet, cleaned with acetone, via a 3M spray gun. The paint is applied in two coats, separated by approximately five minutes. Based on analysis on aluminum test samples, it is estimated that the paint layer is 25 to 35 μm thick.

The PSP is illuminated by a Necsel™ Blue DE Laser, with a power of 10 W and output wavelength of 445 nm, which is the preferred excitation wavelength of the Ru(dpp) luminophore. The laser is diffused before reaching the paint with a Thorlabs optical diffuser. The camera utilized to image the PSP is a Phantom Veo 1310 high speed camera recording at 1000 frames per second with a spatial resolution of 960x1280 pixels. Pressure data is simultaneously recorded by the three Kulite xtl-190 pressure sensors [19] at 1000 Hz, time synchronized with the high speed camera through LabVIEW software.

IV. Data Reduction Methodology

The data processing method used in the current study is a combination of techniques from [14, 15, 18, 20, 21]. The process is summarized in the flowchart shown in Fig. 4, and further discussed in the following subsections.

A. Data Acquisition

Data is collected simultaneously from the Phantom camera and three Kulite pressure transducers. The intensity I of the PSP is recorded at each pixel by the Phantom camera, producing an array of data 960 by 1280 by the number of frames, N. The Kulites, after calibration, provided the gauge pressure p_{kul} at their individual locations over time, a matrix of 3 by N. Both data sets were time synchronized with labview software during the collection process.

B. PSP Calculations

This study converts the intensity of the PSP at each individual pixel to pressure utilizing the Stern-Volmer relation, show in Eqn. 1 [14],

$$\frac{I_{ref}}{I} = A + B\frac{p}{p_{ref}} \qquad , \tag{1}$$

where I_{ref} is the reference intensity of the PSP, I is the experimental intensity, p_{ref} and p are the reference and experimental pressures, and A and B are calibration constants. This equation normalizes intensity and pressure, then correlates the two quantities with an inverse linear relationship, where A and B are found through experimental calibration. This method is well established in the use of PSP for its ability to reduce errors due to paint thickness, the camera's viewpoint, and illumination intensity [14, 15]. Interested readers are directed to the work of Gregory et al. [15]. and the book by Liu et al. [14] for more details.

1. PSP Calibration

In order to utilize the Stern-Volmer relation, the constants A and B need to be obtained for the particular PSP recipe utilized. The constants A and B are found experimentally utilizing an a-priori calibration (top left of Fig. 4). The viewing window of a thermal vacuum chamber is coated with this PSP mixture. At a constant, room temperature, this PSP is exposed to a series of pressures below and at atmospheric pressure and viewed through the window, mimicking the back-imaging experimental setup. One pair of pressure/intensity measurements is then selected as a "reference". $\frac{I_{ref}}{I}$ and $\frac{p}{p_{ref}}$ is calculated for each pressure tested. By applying a linear fit, the calibration constant A and B are found. The resulting calibration is shown graphically in Fig. 5.

The I_{ref} used during the ground-effect experiments was the intensity emitted by the PSP in a "propeller off" situation (top right of Fig. 4) before the day of testing. Assuming this condition is p_{ref}, zero gauge pressure, this provides a matrix 960 by 1280 in size. This data is filtered spatially by averaging over the nearest 31 by 31 pixels, approximately 14.3 by 14.3 mm, mimicking the area of the Kulite pressure transducer head. This reduces the noise of each individual pixel and provides a better comparison between the Kulite and PSP pressure measurements.

This provides all the necessary information needed to convert the PSP intensity measurements to pressure. Although all PSP has an inherent temperature bias, the temperature of the air during these tests is expected to be nearly constant and therefore a temperature correction was not conducted.

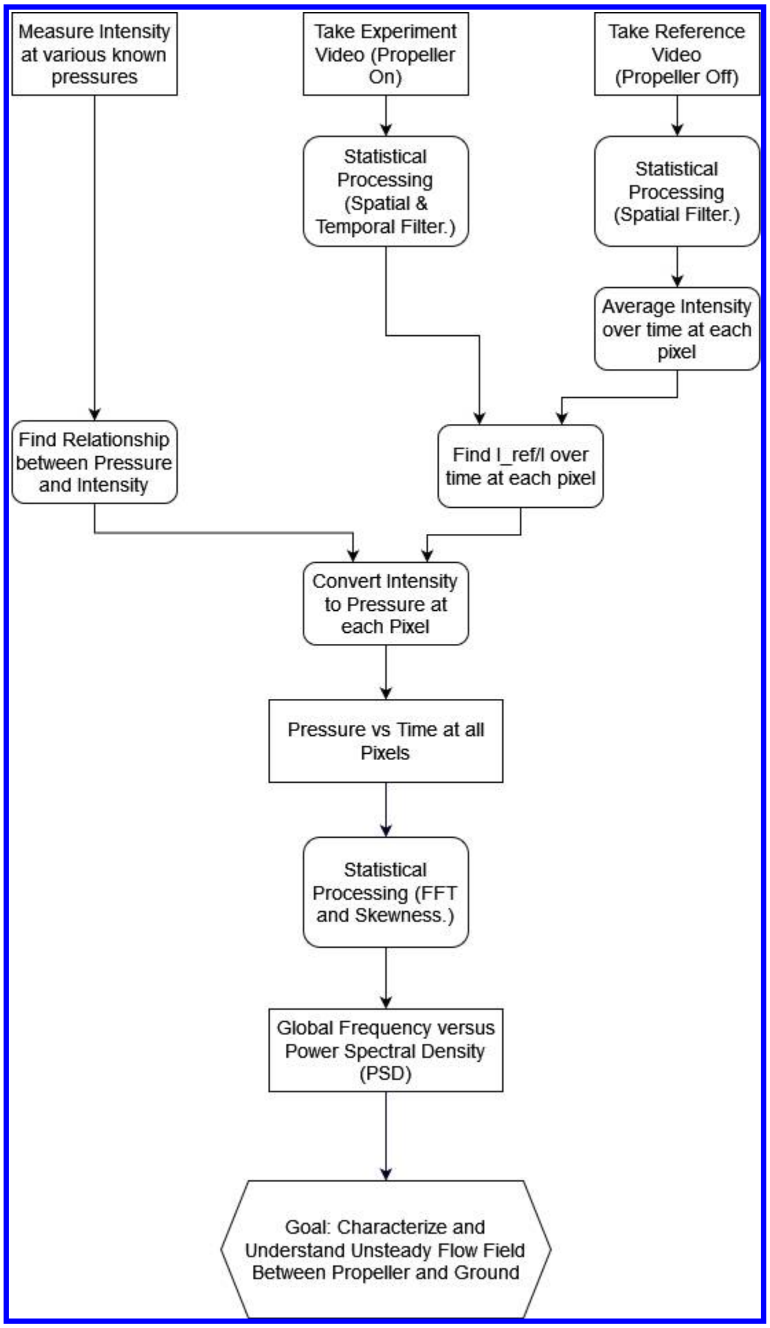

Fig. 4 Flowchart showing PC-PSP data processing method utilized in this study, based on [14, 15, 18, 20, 21]

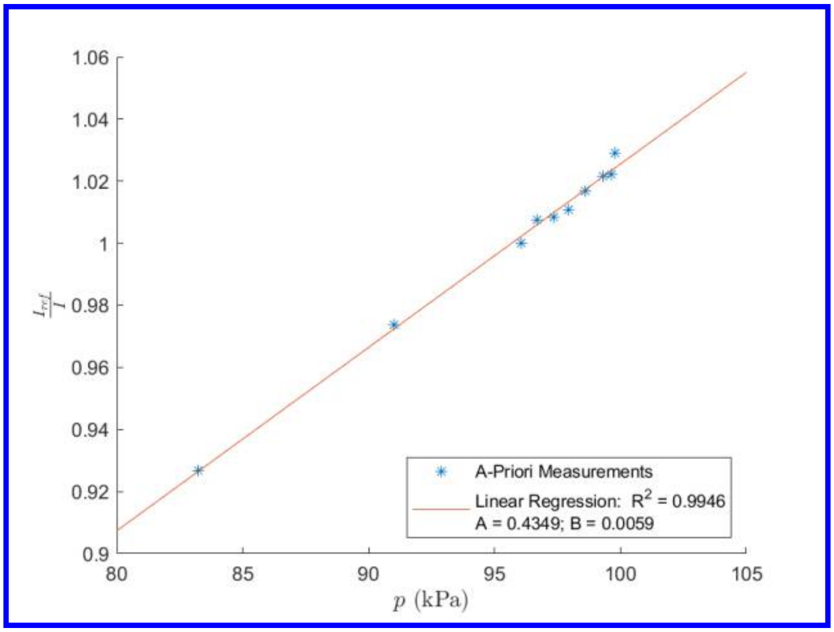

Fig. 5 Graph showing the a-priori $\frac{I_{ref}}{I}$ versus p data and the linear fit applied to convert intensity to pressure.

2. Surface Pressure

The images recorded by the high speed camera produces a 960 by 1280 by N matrix of intensity values. This matrix is filtered spatially over the nearest 31 by 31 pixels, similar to the reference intensity data. In addition, it is temporally filtered by taking a time average over 3 frames (3 milliseconds) at each pixel. Both of these measures are used to reduce noise in the data, but were not found to alter the primary results. The pressure p_{psp} is then found at each pixel utilizing the calibration constants and solving the Stern-Volmer relation, producing a matrix 960 by 1280 by N number of frames. This pressure is then split into a steady and fluctuating component, as shown in Eqn. 2,

$$p_{PSP} = \bar{p}_{PSP} + p'_{PSP} \quad , \tag{2}$$

where p_{PSP} is the pressure calculated from the PSP at each pixel, $\bar{p}_{PSP}$ is the average pressure at each pixel over the testing time, and p'_{PSP} is the fluctuating component of the pressure. By subtracting $\bar{p}_{PSP}$ from p_{PSP}, the unsteady aspects of the flow are analyzed. This is similar to analysis performed by Running et al. [21] to analyze unsteady flows.

3. Fast Fourier Transform

To analyze the unsteady nature of the flow under the rotor blade, the Fast Fourier Transform (FFT) is taken of $\frac{p'}{\bar{p}}$ at each point in the PSP pressure data resulting in the Power Spectral Density (PSD) having units of (1/Hz). This provides the PSD versus frequency, revealing the strength of the unsteady flow at various frequencies in the flowfield globally. This data herein is displayed as either the maximum PSD intensity, the frequency at which this maximum occurs, or a relationship between the two at a given point. It can be shown in a spectral image (PSD vs frequency for a point), contour (image showing PSD or frequency value at each point) or profile (line showing PSD or frequency value along a specified axis).

V. Results and Discussion

A. Surface Pressure Time Trace

Figure 6 shows a direct comparison between the pressure recorded by a Kulite pressure sensor and the pressure recorded by a subsection of the PSP very near this sensor. The specific location of the Kulite and this location in the PSP is shown in Fig. 7. Recall, the PSP data were spatially filtered by the same size as the individual Kulite sensing head. As seen in the figure, both the PSP and the Kulite are fluctuating at what qualitatively appears to be a similar period.

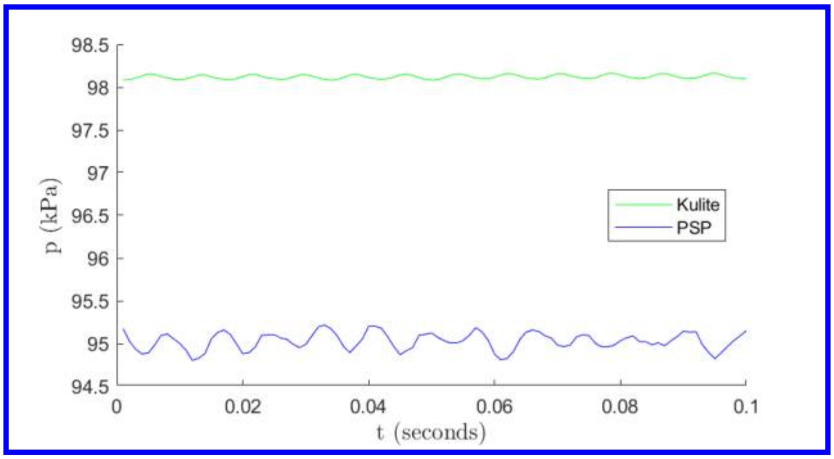

Fig. 6 Pressure data versus time collected from the Kulite pressure sensor farthest from the propeller center compared to the PSP pressure left of the Kulite, spatially filtered over 31 by 31 pixels (approximately 14.3 by 14.3 mm) and temporally filtered over 3 frames (3 milliseconds).

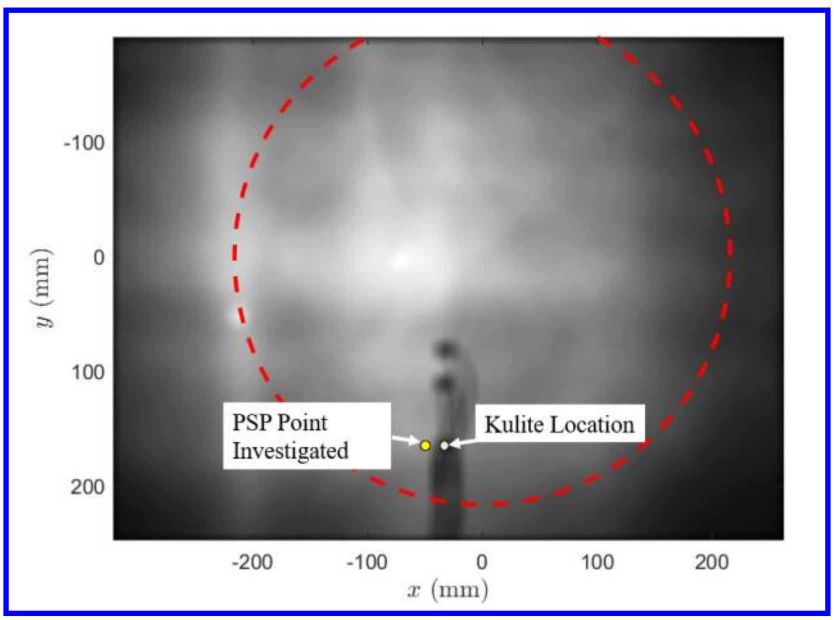

Fig. 7 Location of Kulite and PSP point used to create Fig. 6 plotted on an image of I_{ref}. The Kulites and wiring can be seen in the center bottom of the image, and the point in the PSP data shown with a blue star to the left of the kulite at $y = 175$ mm.

There is an offset of approximately 3% between the mean absolute pressure recorded by the Kulite and the mean absolute pressure recorded by the PSP. It is hypothesized that this difference is due to the light intensity degradation of the PSP between the reference condition, the a-priori calibration, and this particular test. This hypothesis is supported by a discernible decrease in the PSP I from test to test. In addition, the fluctuation amplitude in the PSP signal is approximately 6 times larger than the pressure fluctuation measured by the Kulite. It is similarly hypothesized that this is could be due to light intensity degradation of the PSP affecting the slope of the a-priori calibration (the constant A in Eqn. 1). In addition, it is possible that the propeller blade is visible to the camera through the acrylic plate and PSP layer, therefore biasing the measured PSP I. This hypothesis is discussed in further detail below. The surface pressure oscillations between the Kulite and PSP also appear to be approximately 3/4 of a cycle out of phase. This is likely due to a constant time offset between the Kulite sensor and camera, and will be further analyzed in future testing.

These inconsistencies, in addition to noise in the data, made direct visualization of the absolute pressure field challenging. Therefore, this work focused on the spectral content of the unsteady PSP signal for analysis since this unsteady content is less effected by the degradation of the measured I.

B. PSD of Unsteady Surface Pressure: PSP and Kulite Comparison

This unsteady analysis begins with the spectra of the unsteady surface pressure, shown in Fig. 8. Note, this comes from the same data provided in Fig. 6.

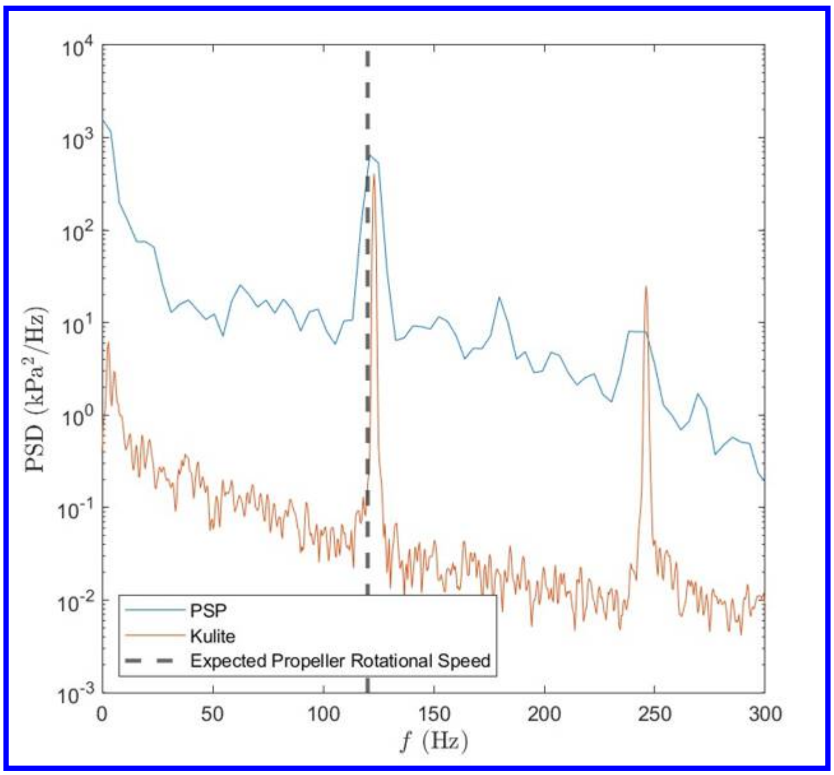

Fig. 8 Comparison of Power Spectral Density (PSD) of fluctuating pressure versus frequency between Kulite sensor and PSP immediately left of Kulite, spatially filtered over 31 by 31 pixels (approximately 14.3 by 14.3 mm) and temporally filtered over 3 frames (3 milliseconds).

To obtain this result, the fluctuating component of the pressure (mathematically, P'_{PSP} as defined in Eqn. 2) is analyzed. This result is then passed to the FFT to produce the content shown in Fig. 8.

Figure 8 shows a primary peak in PSD at 123 Hz in both the PSP and the Kulite data. These peaks show good agreement in frequency and magnitude. There is also agreement between the second amplitude peak located at 246 Hz, where a harmonic is expected. These both correspond with the expected frequency of oscillation in the flowfield, based on the propeller rotating at approximately 60 revolutions per second (rps) and having two blades (therefore, the conditions of the propeller repeat at a rate twice the propeller speed, or approximately 120 Hz). Although the primary and secondary unsteady behavior show good agreement between the Kulite and PSP, the remainder of the spectra does not show good agreement in amplitude. It is hypothesized this is due to the noise in the PSP data, but the exact reason is still unknown. This will be a focus of future follow-on work.

C. Global PSD

The strong agreement in PSD at the peak frequencies of the unsteady phenomenon suggest that the PSP is able to respond to the aerodynamic unsteadiness in the flowfield. This allows the global aspect of the measurement technique to be utilized, as shown by the PSD contour in Fig. 9, where $x = 0$ mm and $y = 0$ mm at the center location of the rotor blade.

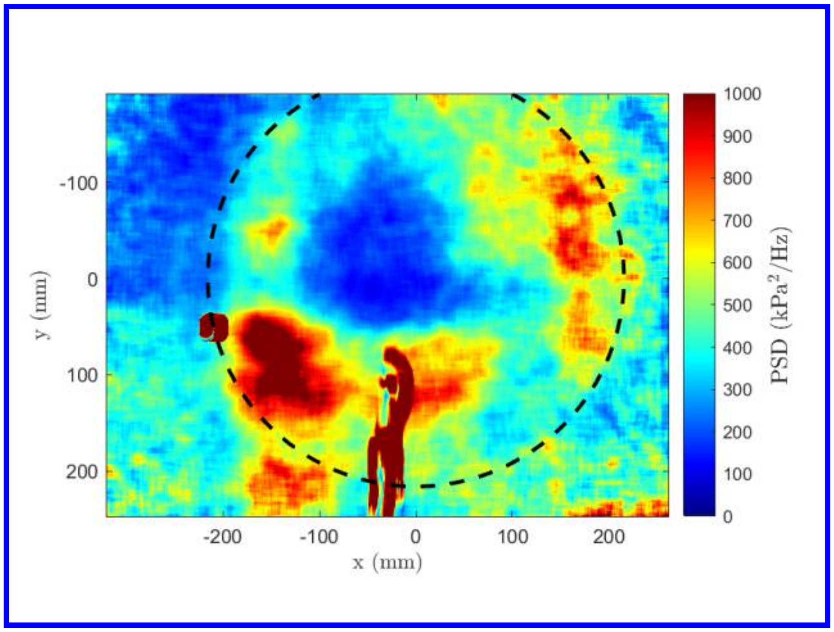

Fig. 9 Contour of PSD amplitude of $\frac{p'_{PSP}}{\bar{p}_{PSP}}$ over the test plate. The deep red sections in the lower center of the image are the Kulite sensors and wiring.

This figure is produced by repeating the process utilized to create Fig. 8 at every point in the recorded image. Then, the maximum PSD intensity, corresponding to the primary peak, is recorded and plotted. This corresponds to reading the maximum PSD value of Fig. 8 (in this case, at 123 Hz) and plotting that value at the relative spatial location the FFT is taken. Note, the higher amplitude, low frequency PSD content observed in Fig. 8 is negated in this analysis. In addition, a circle is plotted on this figure to represent the outer radius of the propeller. This is found by recording images with the PSP and acrylic removed from the test setup, and recording the center and radius of the propeller (Fig. 3).

Figure 9 shows that the PSD has the greatest magnitude in a ring surrounding the propeller. This aligns well with the results from Cai et al. [9]. In that work, radial flow was seen greater than approximately 50% the radius of the propeller, while tangential flow was seen at less than this radius on the ground plate (when the propeller was within $\frac{h}{d}$ <= 0.3). This is qualitatively shown by the tufts, smoke visualization, and oil flow in Fig. 10. It was hypothesized in that work that this observation corresponds to a dividing streamline between tangential and radial flow at the ground plate [9]. The inner radius of the ring observed in Fig. 9 aligns with this hypothesized dividing streamline, at approximately 50% the radius of the propeller. This suggests that the greatest unsteady effect and the highest energy flow occurs beyond approximately 50% the radius of the propeller in the radial flowfield, while the tangential, inner flowfield does not exhibit significant amplitudes of unsteady surface pressure fluctuations, comparatively.

This result is quantitatively better shown in Fig. 11. This figure is created by plotting the average PSD value for a given radius away from the center of the propeller. The blue line corresponds to 50% of the propeller radius, where the dividing streamline is expected [9]. Although there are slight asymmetries observed, it is worth noting that the minimum PSD lies at the location corresponding to the center of the rotor blade ($r = 0$ mm), as expected. In addition, the concavity of the PSD growth outward from this center flips sign very near the location of the previously-observed dividing streamline (i.e., the blue vertical dashed lines). The PSD value is relatively low for $r < 105$ mm, indicating that

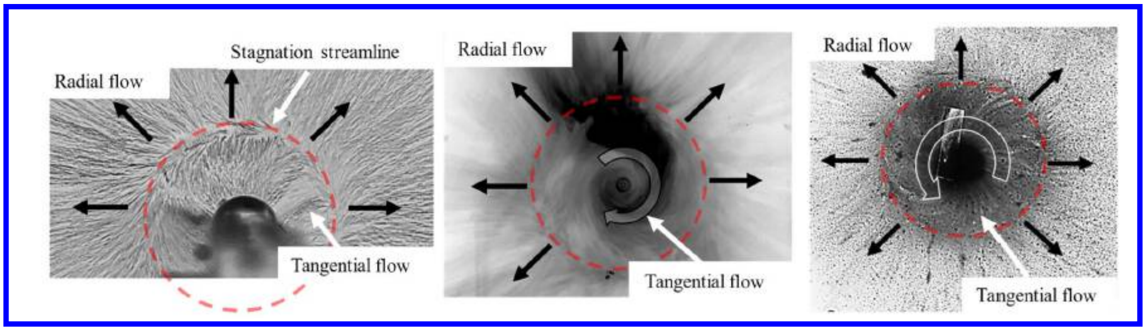

Fig. 10 Tuft (left), smoke (middle), and oil (right) visualization of near-surface flowfield, $\frac{h}{d} = 0.2$, reproduced from [9].

the unsteady surface pressure fluctuations are relatively weak within this tangential-flow core. However, outside of this tangential-flow core, the radial-flow portion has higher PSD, with the local maximum PSD peaking around $r = 175$ mm, and then decreasing, before spiking again. This strong spike likely corresponds to the tip vortices produced as the propeller blades pass over the ground.

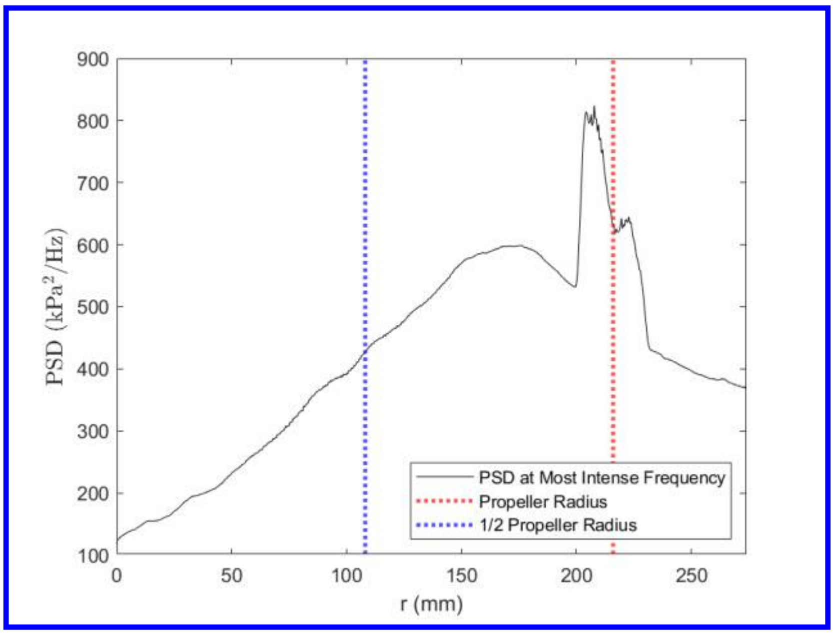

Fig. 11 Radial profile of PSD primary-peak amplitude of $\frac{p'_{PSP}}{\bar{p}_{PSP}}$ taken as an average of rays extending from $y = 0$ mm, $x = 0$ mm in Fig. 9

In conjunction with this information, the frequency at which the PSD recorded the maximum value shown in Fig. 9 and 11 is graphed as a separate contour, shown in Fig. 12. To create this figure, the PSD was calculated at every pixel, and the frequency corresponding to the primary fundamental peak associated with the aerodynamic effect of the propeller was tabulated. See Fig. 8 above for a representative PSD at a single pixel location. Figure 12 reveals that, throughout the flowfield, the unsteady fluctuations occur at a frequency nearly identical to two times the propeller rotational speed.

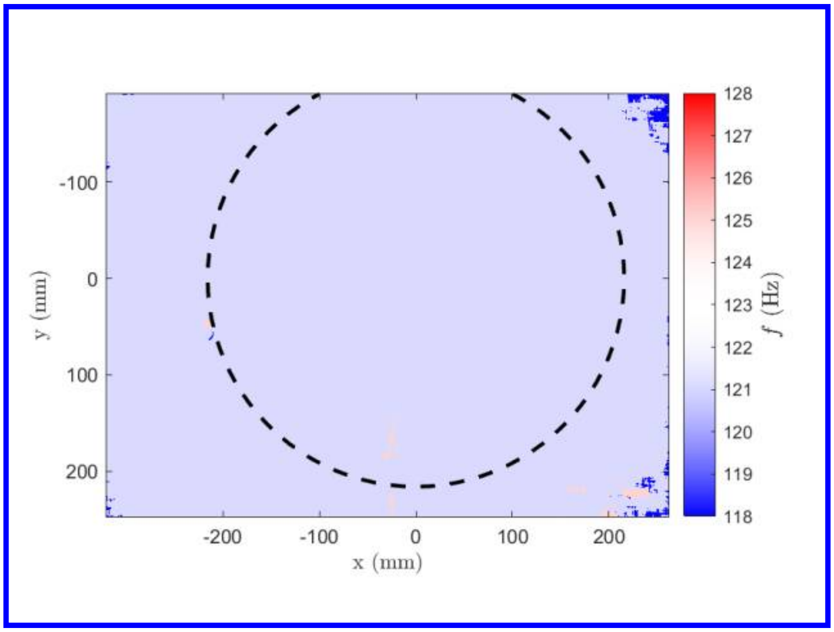

Fig. 12 Contour of frequency at which maximum PSD intensity peak of $\frac{p'_{PSP}}{\bar{p}_{PSP}}$ occurs.

D. Optical Propeller Interference

Figure 12 returns discussion to whether or not this unsteady oscillation shown in the PSP is strictly an observation of the flowfield. It can be argued that the propeller is visible through the paint layer and acrylic, thereby changing the observed intensity of the PSP and influencing the fluctuating pressure component p'_{PSP}. The increased magnitude in fluctuations of the PSP relative to the Kulite, as seen in Fig. 6 could be argued to be due to this effect. If the camera could "see" the propeller through the PSP and acrylic layers, it would cause the recorded intensity at each pixel within the propeller radius to fluctuate as it rotates. Alternatively, the laser could reflect off the propeller and create a region of higher lumination under the propeller. Either of these effects would cause an oscillation in p'_{PSP} at a different magnitude than the Kulite recorded pressure, much like that seen in Fig. 6. There is no definitive evidence against this argument. However, as seen in Fig. 12, the PSP is capable of measuring surface pressure oscillations outside of the propeller radius at the same frequency as within the propeller radius. This confirms that the PSP is indeed responding to the aerodynamic effects independent of the potentially visible blade, since outside of the blade radius, the blade is obviously not visible. Therefore, although it cannot be ruled out that the propeller is not *influencing* the results, it is hypothesized that the PSP is primarily recording the effects of the pressure field. Confirming this hypothesis will also be a focus of future follow-on work.

VI. Conclusions

This study utilized back-imaged pressure-sensitive paint (PSP) to analyze the pressure field under a small propeller blade, as would be utilized in unmanned aerial vehicles or small rotorcraft. Through this technique, the global, unsteady surface pressure fluctuations pertaining to the flow were able to be analyzed.

By globally viewing the frequency at which the maximum power spectral density (PSD) amplitude occurred, it was seen that throughout the pressure field the strongest fluctuating component occurred at 123 Hz, or approximately two times the expected rotational speed of the propeller. This corresponds with the hypothesized oscillation of the flowfield for a two-blade propeller. This PSD was also in good agreement with the Kulite measurements. In viewing the global PSD, a clear, low amplitude center is surrounded by a ring of maximum amplitude at approximately 75% the radius of the propeller. In addition, by viewing the PSD amplitude as an average profile along the propeller radius, the concavity of this spectral amplitude profile changed signs near the blade-radius percent where previous works by the authors qualitatively measured a dividing streamline between primarily tangential flow and radial flow [9].

There remains unanswered questions as to the differences in amplitude of the PSD between the PSP and the Kulite pressure transducers in the flow as well as the potential that the propeller itself is optically influencing the measurement technique. These topics will be the focus of future follow-on work. Despite these questions, it was determined the PSP was able to successfully image the pressure field and provide global, unsteady analysis of the flow. These results show promise in the use of PSP to quantitatively analyze the flowfield under small rotorcraft blades in future work.

VII. Acknowledgements

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References

[1] Betz, A., "The Ground Effect on Lifting Propellers," Tech. Rep. 836, NACA, 1937.

[2] Griffiths, D. A., Ananthan, S., and Leishman, J. G., "Predictions of Rotor Performance in Ground Effect Using a Free-Vortex Wake Model," *Journal of the American Helicopter Society*, Vol. 50, No. 4, 2023, pp. 302–314.

[3] Cheeseman, C., and Bennett, W., "The effect of the Ground on a Helicopter Rotor in Forward Flight," Tech. Rep. 3021, British R M, 1957.

[4] Zbrozek, J., "Ground Effect on the Lifting Rotor," Tech. Rep. 2347, British R M, 1947.

[5] Fradenburgh, E., "The Helicopter as a Ground Effect Machine," *Journal of the American Helicopter Society*, 1960, pp. 24–33.

[6] Knight, M., and Hefner, R., "Analysis of Ground Effect on the Lifting Airscrew," Tech. Rep. TN-835, NACA, 1941.

[7] Hayden, S., J., "Effect of the ground on helicopter hovering power required." Proceedings of the AHS 32nd Forum, 1976.

[8] Cai, G. S. A. A., J., and OL, V., M., "Propeller Ground and Ceiling Effect Parametric Data," *AIAA Journal of Aircraft*, Vol. 58, No. 3, 2020, pp. 700–703. https://doi.org/10.2514/1.C035839.

[9] Cai, G. S., J., and OL, V., M., "Effect of Partial Ground and Partial Ceiling on Propeller Performance," *AIAA Journal of Aircraft*, 2022. https://doi.org/10.2514/1.C036974.

[10] Dekker, R. D. B. J. S. F., H., and Tuinstra, M., "Aerodynamic Interactions of Side-by-Side Rotors in Ground Proximity," *AIAA Journal*, Vol. 60, No. 7, 2022. https://doi.org/10.2514/1.J061105.

[11] He, L. K., X., "A New Quasi-Steady In-Ground Effect Model for Rotorcraft Unmanned Aerial Vehicles," ASME 2019 Dynamic Systems and Control Conference, DSCC 2019-9025, 2019. https://doi.org/10.1115/DSCC2019-9025.

[12] Sanchez-Cuevas, M. V. H. G., P., and Ollero, A., "Aerodynamic Effects in Multirotors Flying close to Obstacles: Modelling and Mapping," Fourth Iberian Robotics conference, pp. 63-74, 2020. https://doi.org/10.1007/978-3-030-35990-4_6.

[13] Calvert, M., and Wenren, Y., "Rotorcraft Outwash and Human Stability," AIAA Scitech 2019, San Diego, CA, Jan 2019, AIAA 2019-1095., 2019.

[14] Liu, T., Sullivan, J. P., Asai, K., Klein, C., and Egami, Y., *Pressure and temperature sensitive paints*, 2^{nd} ed., Springer, ewerbestrasse 11, 6330 Cham, Swizterland, 2021.

[15] Gregory, J. W., Sakaue, H., Liu, T., and Sullivan, J. P., "Fast Pressure-Sensitive Paint for Flow and Acoustic Diagnostics," *Annual Review of Fluid Mechanics*, Vol. 46, No. 1, 2014, pp. 303–330. https://doi.org/10.1146/annurev-fluid-010313-141304.

[16] Turpin, A. M., Granlund, K. O., Hayashi, T., and Sakaue, H., "Back-imaging of polymer-ceramic pressure-sensitive paint," *Measurement Science and Technology*, Vol. 32, No. 10, 2021, p. 104008. https://doi.org/10.1088/1361-6501/ac0a0f.

[17] "APC Propellers | Quality Propellers that are Competition Proven," , 2023. URL https://www.apcprop.com/.

[18] Sakaue, H., Kakisako, T., and Ishikawa, H., "Characterization and Optimization of Polymer-Ceramic Pressure-Sensitive Paint by Controlling Polymer Content," *Sensors*, Vol. 11, No. 7, 2011, pp. 6967–6977. https://doi.org/10.3390/s110706967.

[19] Kulite, "XTEL-10L-190 Pressure Transducers," , 2023. URL http://www.kulite.com.

[20] Zare-Behtash, H., Lo, K., Yang, L., and Kontis, K., "Pressure sensitive paint measurements at high Mach numbers," *Flow Measurement and Instrumentation*, Vol. 52, 2016, pp. 10–16. https://doi.org/10.1016/j.flowmeasinst.2016.02.004.

[21] Running, C. L., and Juliano, T. J., "Global measurements of hypersonic shock-wave/bounday-layer interactions with pressure-sensitive paint," *Experiments in Fluids*, Vol. 62, No. 5, 2021, p. 91.